



AI CAREERS FOR WOMEN

LINKEDIN FELLOWSHIP

PERSONAL FINANCIAL
HEALTH FORECASTING



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Project Guide

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Outline

- Abstract of the Project
- Problem Statement
- Proposed Solution
- System Architecture
- Live Demo of the Project
- Embedded Video of Project
- Conclusion
- Future Scope



Abstract

- FHFM scores 1,000 individuals on financial health (0–100).
- Objective: Forecast 5-year savings & reduce Risky category.
- Tools: Excel, Power BI + Copilot, and Claude AI.
- Outcome: Cut Risky rate from 96.8% via savings strategies.
- Dataset: 1,000 records, 16 variables, ages 21–59.



Problem Statement

- 96.8% of 1,000 individuals fall in the Risky category.
- Income alone does not guarantee financial health.
- Affects individuals, advisors & fintech platforms.
- Existing tools show past data only — no forecasting.
- Need: Integrate Savings, Expense & Loan Ratios into one score.



Proposed Solution

The FHFM assigns each individual a Financial Health Score (0–100) across 4 stages: (1) Data Input — personal records, behavioral history, macro signals (RBI, CPI, NSE). (2) Preprocessing — ratio normalization, feature engineering. (3) Processing — FHS Scoring Engine + XGBoost/LSTM Forecasting + Monte Carlo Simulation (10,000 paths). (4) Output — Power BI dashboard with risk alerts, 5-year savings forecasts, and AI action cards via Copilot and Claude.



System Architecture

Data Input / Source

Excel dataset: 1,000 individuals with income, expenses, savings, loans, investments, emergency fund. External: RBI API, NSE/BSE Market Volatility Index, CPI inflation data.

AI / Tool Integration

Excel: Pivot tables + advanced formulas. Power BI + Copilot: DAX measures, NL queries, automated insights. Claude AI: Scenario narratives and prompt-based recommendations.

Processing / Analysis

Missing value imputation, normalization (0-1 scale), ratio engineering: Savings Ratio, Expense Ratio, Loan Ratio, outlier capping and rolling average smoothing.

Output / Dashboard

Dynamic FHS Dashboard, Scenario Forecast Charts, Risk Flag Alerts, Priority Action Cards, Goal Readiness Tracker, and Shock Impact Reports.

Live Demo of Project

Here's the list of some demo screenshots / features of the project:



FHS Dashboard

Power BI dashboard with FHS score 0-100 per individual, color-coded by risk: Risky (red), Average (amber), Good (green).



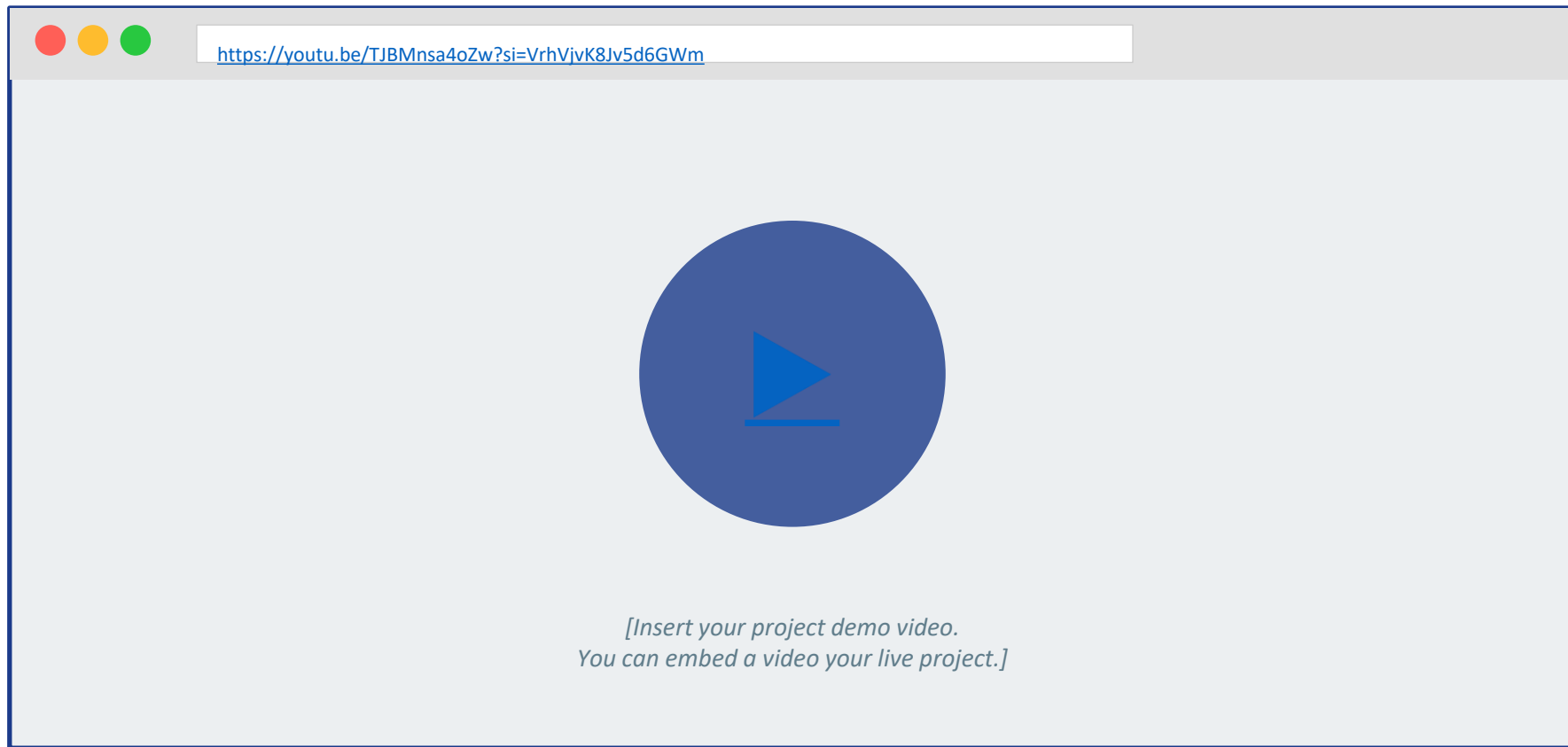
5-Year Forecast

XGBoost + LSTM
comparing
Future_Savings_5Y vs
Improved_Savings_5Y —
50% savings gain with
behavioral change.



Monte Carlo Sim

10,000 Monte Carlo
paths stress-testing job
loss, salary hike, EMI
payoff and CPI shock
with probability
outcomes.



Conclusion

FHFM analyzed 1,000 individuals — 96.8% in Risky category. Reducing expense ratio by 10% improves 5-year savings by up to 50%. All objectives achieved: EDA, data cleaning, forecasting model (XGBoost + LSTM), Power BI dashboard, Monte Carlo simulation, and full AI integration with Copilot and Claude.



Future Scope

Phase 2: Integrate live RBI/NSE APIs for real-time rates. Build Fintech REST API with push risk alerts. Add behavioral ML for spending pattern detection. Scale to 100K+ users for regional insights. Launch mobile app with Claude AI for real-time personalized financial coaching.



Thank You

AICW Fellowship | Student Project Presentation 2026

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